

CSE 432 Robotics

VELOCITY KINEMATICS- Singularity

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SINGULARITIES

- Identifying singularities is important because they affect manipulator performance and control.
- Singularities correspond to configurations where certain directions of motion become unattainable.
- At a singularity, **bounded end-effector velocities** may require **unbounded joint velocities**.
- Likewise, **bounded end-effector forces/torques** may require **unbounded joint torques**.
- Singularities often (but not always) lie on the **boundary of the workspace**, typically at points of maximum reach.



SINGULARITIES

- Singularities may correspond to workspace points that become **unreachable** under small variations in link parameters (e.g., length, offsets).
- Near a singularity, the **inverse kinematics may not have a unique solution**:
 - There may be **no solution**.
 - Or there may be **infinitely many solutions**.



SINGULARITIES

- The $6 \times n$ Jacobian $J(q)$ maps joint velocities $\dot{q}$ to end-effector velocities

$$\xi = (v, \omega), \xi = J(q) \dot{q}.$$

- For a two-link planar arm, the Jacobian has **two columns**, but $\xi \in \mathbb{R}^6$,
- Thus, **six linearly independent columns** are required for the end-effector to achieve any arbitrary spatial velocity.



SINGULARITIES

- The **rank** of a matrix equals the number of **linearly independent** rows or columns.
- When $\text{rank } J = 6$, the end-effector can execute any arbitrary velocity.
- For a matrix $J \in \mathbb{R}^{6 \times n}$, the $\text{rank } J = \min(6, n)$.
- For a two-link planar arm,
$$\text{rank}(J) \leq 2,$$
- because the Jacobian has only two columns.



SINGULARITIES

- For an anthropomorphic arm with a spherical wrist,
 $\text{rank}(J) \leq 6$,
- matching the six possible end-effector velocity components.
- The **rank of a Jacobian is not constant**; it can change depending on the robot configuration q .
- The rank of $J(q)$ determines how many independent motion directions the manipulator can generate at a given configuration.
- A configuration is **singular** when
 $\text{rank}(J(q)) < \text{rank}_{\max}$.



SINGULARITIES

- Several methods can be used to determine Jacobian singularities.
- A square matrix is singular when its **determinant is zero**:
$$\det J(q) = 0.$$
- In practice, solving
$$\det J(q) = 0$$
- is often difficult because the equation is nonlinear.
- For manipulators with a **spherical wrist**, the **decoupling method** provides a simpler way to analyze singularities.



Decoupling of Singularities

- Suppose that $n = 6$, that is, the manipulator consists of a 3-DOF arm with a 3-DOF spherical wrist.
- In this case the Jacobian is a 6×6 matrix, and a configuration $\mathbf{q}$ is singular if and only if

$$\det(J(\mathbf{q})) = 0$$

- If we now partition the Jacobian J into 3×3 blocks as

$$J = [J_P \mid J_O] = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$



Decoupling of Singularities

- Since the final three joints are always revolute

$$J_o = \begin{bmatrix} z_3 \times (o_6 - o_3) & z_4 \times (o_6 - o_4) & z_5 \times (o_6 - o_5) \\ z_3 & z_4 & z_5 \end{bmatrix}$$

- As singularities are typical of the mechanical structure and do not depend on the frames chosen to describe kinematics, it is convenient to choose the origin of the end-effector frame at the intersection of the wrist axes.
- The wrist axes intersect at a common point o , that $o_3 = o_4 = o_5 = o_6 = o$, then J_o becomes

$$J_o = \begin{bmatrix} 0 & 0 & 0 \\ z_3 & z_4 & z_5 \end{bmatrix}$$



Decoupling of Singularities

- In this case the Jacobian matrix has the block triangular form

$$J = \begin{bmatrix} J_{11} & 0 \\ J_{21} & J_{22} \end{bmatrix}$$

- with determinant

$$\det(J) = \det(J_{11}) \det(J_{22})$$



Decoupling of Singularities

- The full manipulator is singular when either:

- the **arm Jacobian** satisfies

$$\det(J_{11}) = 0,$$

- or the **wrist Jacobian** satisfies

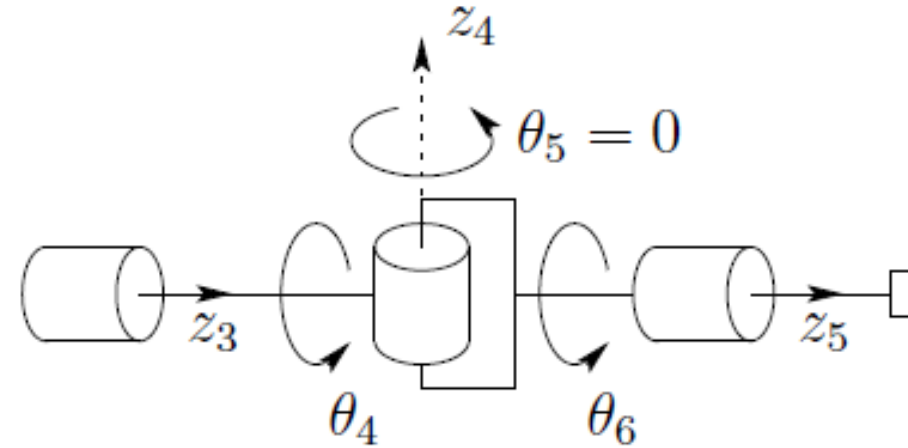
$$\det(J_{22}) = 0.$$

- Therefore, the complete set of singular configurations is the **union** of:

- arm singularities ($\det J_{11} = 0$), and
- wrist singularities ($\det J_{22} = 0$).



Wrist Singularities



- A spherical wrist is in a singular configuration whenever the vectors z_3 , z_4 and z_5 are linearly dependent.
- The only singularity of the spherical wrist occur when the joint axes z_3 and z_5 are collinear.
- It is unavoidable without imposing mechanical limits on the wrist design to restrict its motion in such a way that z_3 and z_5 are prevented from lining up.



Arm Singularities

- To investigate arm singularities, we need only to compute J_{11} but with the wrist center o in place of o_n .



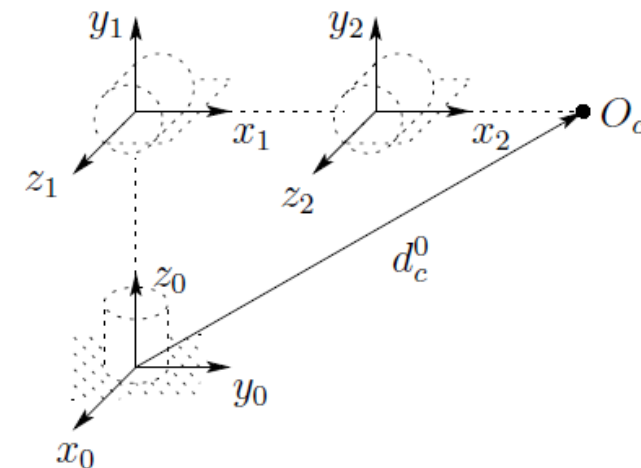
Elbow Manipulator Singularities

- Consider the three-link articulated manipulator with coordinate frames attached as shown below.
- We can easily obtain J_{11} as follows:

$$J_{11} = \begin{bmatrix} -a_2 s_1 c_2 - a_3 s_1 c_{23} & -a_2 s_2 c_1 - a_3 s_{23} c_1 & -a_3 c_1 s_{23} \\ a_2 c_1 c_2 + a_3 c_1 c_{23} & -a_2 s_1 s_2 - a_3 s_1 s_{23} & -a_3 s_1 s_{23} \\ 0 & a_2 c_2 + a_3 c_{23} & a_3 c_{23} \end{bmatrix}$$

The determinant of J_{11} is

$$\det J_{11} = a_2 a_3 s_3 (a_2 c_2 + a_3 c_{23})$$





Elbow Manipulator Singularities

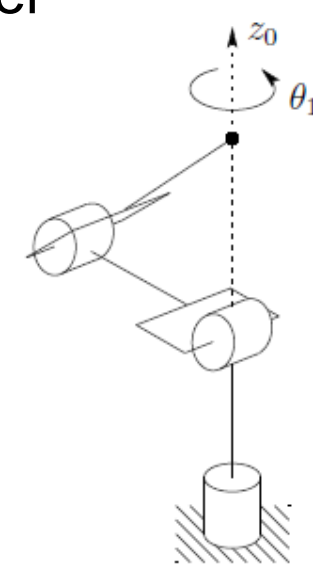
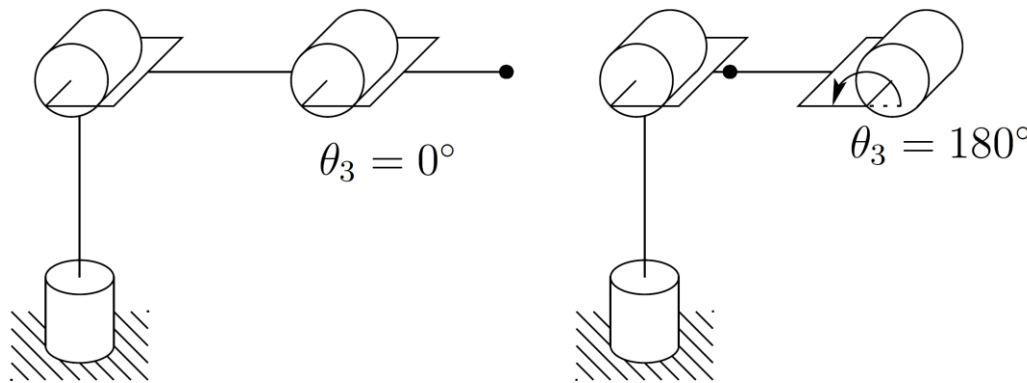
- The elbow manipulator is in a singular configuration whenever

$$s_3 = 0 \rightarrow \theta_3 = 0 \text{ or } \pi$$

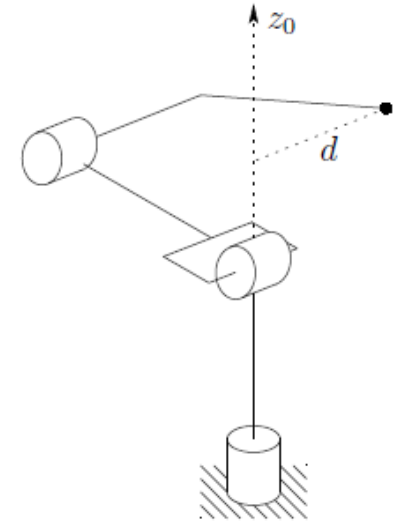
And whenever

$$a_2 c_2 + a_3 c_{23} = 0$$

- Which occurs whenever the wrist center is along z_0 axis.



Singularity of the elbow manipulator with no offsets



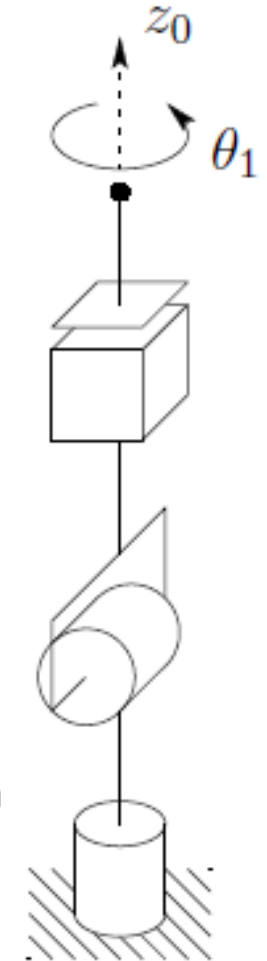
Elbow manipulator with shoulder offset



Spherical Manipulator

- This manipulator is in a singular configuration when the wrist center intersects z_0 as shown since, as before, any rotation about the base leaves this point fixed

Singularity of
spherical
manipulator with
no offsets

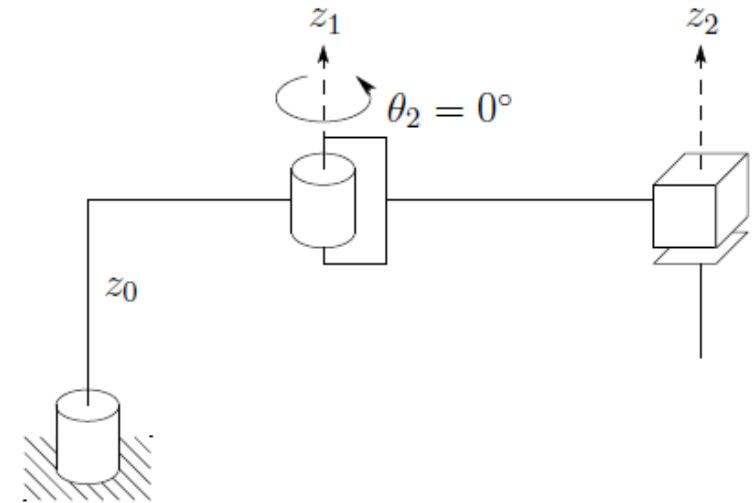




SCARA Manipulator

- The complete Jacobian for the SCARA manipulator is as follows:

$$J = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} & 0 & 0 \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$



- This Jacobian is simple enough to be used directly rather than deriving the modified Jacobian as we have done before.
- Geometrically the only singularity occurs when the elbow is fully extended or fully retracted.



SCARA Manipulator

- The portion of the Jacobian of the SCARA governing arm singularities.

$$\begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} & 0 & 0 \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \\ \dot{\theta}_4 \end{bmatrix}$$



SCARA Manipulator

The portion of the Jacobian governing arm singularities is

$$J_{11} = \begin{bmatrix} \alpha_1 & \alpha_3 & 0 \\ \alpha_2 & \alpha_4 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

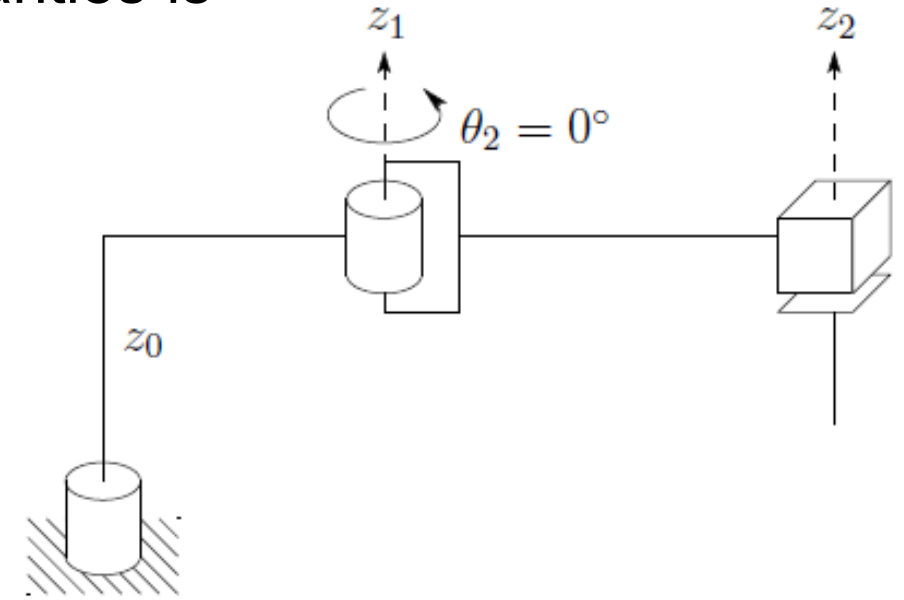
Where

$$\alpha_1 = -a_1 s_1 - a_2 s_{12}$$

$$\alpha_2 = a_1 c_1 + a_2 c_{12}$$

$$\alpha_3 = -a_1 s_{12}$$

$$\alpha_4 = a_1 c_{12}$$



- The rank of J_{11} will be less than three precisely whenever $\alpha_2 \alpha_3 = 0$
- It is equivalent to

$$s_2 = 0 \rightarrow \theta_2 = 0 \text{ or } \pi$$



Static Force/Torque Relationships

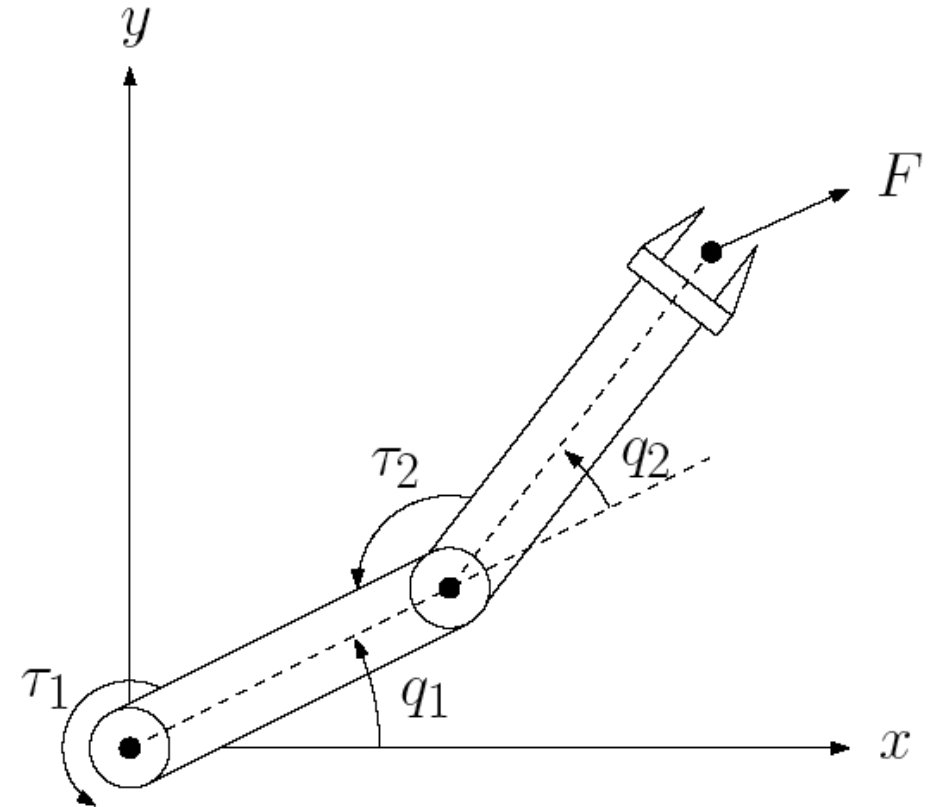
- We are interested in expressing the relationship between the joint torques and the forces and moments at the end effector
 - Important for dynamics, force control, etc
- Let the vector of forces and moments at the end effector be represented as: $F = [F_x \quad F_y \quad F_z \quad n_x \quad n_y \quad n_z]^T$
- Then we can express the joint torques, τ , as:
$$\tau = J^T(\mathbf{q})F$$
- The formulation can be easily derived using the principle of virtual work.



Force/torque relationships

- Example: for a force F applied to the end of a planar two-link manipulator, what are the resulting joint torques?
 - First, remember that the Jacobian is:

$$J = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

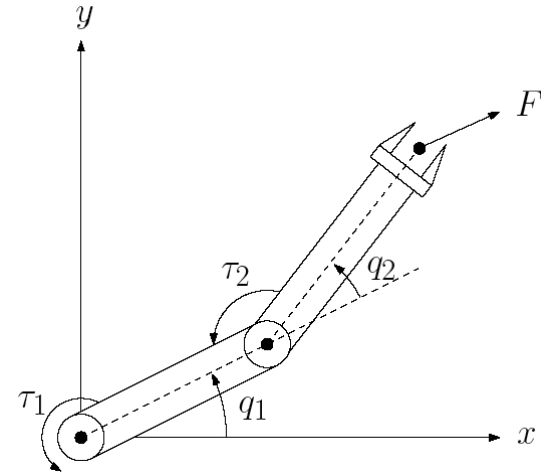




Force/torque relationships

- Example: for a force F applied to the end of a planar two-link manipulator, what are the resulting joint torques?
 - Thus, the joint torques are:

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & a_1 c_1 + a_2 c_{12} & 0 & 0 & 0 & 1 \\ -a_2 s_{12} & a_2 c_{12} & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \\ n_x \\ n_y \\ n_z \end{bmatrix}$$
$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} F_x(-a_1 s_1 - a_2 s_{12}) + F_y(a_1 c_1 + a_2 c_{12}) + n_z \\ F_x(-a_2 s_{12}) + F_y(a_2 c_{12}) + n_z \end{bmatrix}$$





INVERSE VELOCITY

- The Jacobian relationship

$$\xi = J\dot{q}$$

- When the Jacobian is square and nonsingular

$$\dot{q} = J^{-1}\xi$$

- For manipulators that do not have exactly six links ($n > 6$)

$$\dot{q} = J^+\xi + (I - J^+J)b$$

in which $b \in \mathbb{R}^n$ is an arbitrary vector and J^+ is the right pseudoinverse of J .



Any Questions